

MMAE 445: Computer-Aided Design and Manufacturing

Syllabus

Spring 2026

Mondays and Wednesdays (CRN: 53242) – [01/12/26 – 05/09/26]

3:15pm – 4:30pm

Main Campus – Stuart Building (SB112F) & Facility Building (FB125)

COURSE DESCRIPTION

This course covers various topics related to the design and manufacturing process using computer-based tools and technologies. The design aspects focus on the techniques and applications used for creating geometry-driven engineering documents. While the manufacturing portion explores the capabilities of simulating machining centers and generating their toolpath operations. Students will be expected to read and make 2D Blueprints using Geometric Dimensional and Tolerancing (GD&T) techniques that link to 3D Parametric Models derived from Solid, Curve, and Surface geometry. Functional mechanical assemblies are to be digitally constructed and tested against through design optimization practices utilizing Finite Element Analysis (FEA) studies. NC and G-Code toolpaths will then be programmed, generated, and imported into a Computer Numerical Controller (CNC) for 2-Axis, 3-Axis, and 5-Axis machining. This will culminate in a well-rounded understanding of the engineering methodology, design for manufacturability and assembly (DFMA).

COURSE OBJECTIVES & TOPICS

After taking the course the student should be able to:

- Understand the importance of design for manufacturability and assembly (DFMA)
- Understand the principles of geometric dimensioning and tolerancing (GD&T)
- Understand how to make 2D blueprints and 3D models using CAD software
- Understand the applications of sketches, features, constraints, and mates in parts and assemblies
- Understand how to create configuration tables and equations for parts and assemblies
- Understand the impact of materials properties on manufacturing processes and part design
- Understand how to analyze and optimize designs using simulation studies for product performance
- Understand how to generate toolpaths and simulate machining operations using a CAM software
- Understand how to read and write NC and G-Codes programs required for CNC machines
- Understand the different manufacturing machines, cutting tools, and processes in manufacturing
- Understand the fundamentals of CAD and CAM software and their compatibility with each other
- Understand the computer aided tools necessary for the design and manufacturing process

Relationship of Course to ABET Student Outcomes:

ABET Criterion	Program Outcome	Status
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	1
2	An ability to apply engineering design to produce solutions that meets specific needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1
3	An ability to communicate effectively with a range of audiences	0
4	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgements, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	0
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	0
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgement to draw conclusions	0
6.1	Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice	2
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies	1

INTM/MMAE 445: Class Outline (First 8 Weeks)

Day	Content	Topics
Week 1 (M)	Blueprint Reading & GD&T	Title Blocks, Revision Tables Dimensions, Views, GD&T Symbols and Classifications
Week 1 (W)	G-Code Reading & Programming	G-Codes, M-Codes, Lines, Arcs, & Tool Cutter Compensation
Week 2 (M)	HOLIDAY / NO CLASS	Work on Quizzes & Watch Online Lecture
Week 2 (W)	Soft Riser (CAD)	Planes, Constraints, 2D Sketching with Lines and Arcs, & 3D Boss Extrusions and Cuts
Week 3 (M)	Soft Riser (CAD)	Planes, Constraints, 2D Sketching with Lines and Arcs, & 3D Boss Extrusions and Cuts
Week 3 (W)	Soft Riser (CAD)	Offsets, Angles, Trimming, Hole Types and Classifications, Blueprint Making, Sketch Levels, Views, & Dimensions
Week 4 (M)	Soft Riser (CAM)	Levels, Planes, Wireframes, Stock Setup, Tool Libraries, 3-Axis Face Milling, Contours & Holes for First Operation
Week 4 (W)	Soft Riser (CAM)	Machine Setup for Multi-Op Programming, Toolpath Groups, & Dynamic Milling for Second Operation
Week 5 (M) (QUIZ DUE)	Fidget Spinner (CAD)	Insert Part Modeling, Parent-Child References, Mirrored Features, Revolves, & Circular Patterns
Week 5 (W) (QUIZ DUE)	Fidget Spinner (CAD)	3D Assemblies, Foreign vs Native Solids, Basic, Advanced, and Mechanical Mates
Week 6 (M)	Bracket Exercise (CAD)	Sketch Optimization, Shells, Ribs, Typical Walls, and Creating Soft Jaws using 3D Assemblies
Week 6 (W)	Bracket Exercise (CAD)	Revision Management, Design Optimization, & Finite Element Analysis (FEA) Static Studies
Week 7 (M)	Bracket Exercise (CAM)	Importing and Transforming Multiple Components, 3-Axis Pockets and Multi-Op Programming
Week 7 (W)	Bracket Exercise (CAM)	Revision Management & Square vs Bull Endmills
Week 8 (M) (Midterm)	Bracket Exam (CAD) – Q&A	Multi-Stage Revision Design
Week 8 (W) (Midterm)	Bracket Exam (CAM) – Q&A	Multi-Stage Revision Programming

Schedule Continues on Next Page

INTM/MMAE 445: Class Outline (Second 8 Weeks)

Day	Content	Topics
Week 9 (M)	Playing Card Holder (CAD)	Linear Patterns vs Fill Patterns
Week 9 (W)	Playing Card Holder (CAD)	Drafts and Body Combines (Boolean Remove)
Week 10 (M)	Playing Card Holder (CAD) Detergent Bottle (CAD)	FEA Drop Test Studies & Custom Threads
Week 10 (W)	<u>Playing Card Holder (CAM)</u> Detergent Bottle (CAM)	3-Axis Milling for Patterns and Repetitive Hole Making Injection Mold Design
Week 11 (M)	HOLIDAY / NO CLASS	Work on All Exercises & Homework
Week 11 (W)	HOLIDAY / NO CLASS	Work on All Exercises & Homework
Week 12 (M)	<u>Playing Card Holder (CAM)</u> Detergent Bottle (CAM)	3-Axis Surfacing vs 5-Axis Swarfing Injection Mold Design
Week 12 (W)	<u>Playing Card Holder (CAM)</u> Detergent Bottle (CAM)	<u>5-Axis Indexing</u> & <u>Multi-Axis Planes</u> Injection Mold Design
Week 13 (M)	Detergent Bottle (CAD)	Driven Curves & Controlled Lofts
Week 13 (W)	Detergent Bottle (CAD)	Advanced FEA Flow Simulation & Mold Making
WEEK 14 (M)	Detergent Bottle (CAM)	3+2 Roughing and Mirror Patterns for Toolpaths
WEEK 14 (W)	Detergent Bottle (CAM)	Multi-Stage 5-Axis Surface Tracing
WEEK 15 (M)	ARBOR PRESS (CAD) – Q&A	Top-Down Assembly Modeling, & Advanced Surfacing
WEEK 15 (W)	ARBOR PRESS (CAM) – Q&A	Advanced Surfacing & Complex Work Holding
Week 16 (M) (Final)	ARBOR PRESS (CAD) – Q&A	Finish All Exercises, Homework, and Final Exam
Week 16 (W) (Final)	ARBOR PRESS (CAM) – Q&A	Finish All Exercises, Homework, and Final Exam

[MMAE 445](#)

YouTube: <https://www.youtube.com/@artofthepart/playlists>

CLASS FORMAT

Due to the hands-on nature of this class, the primary delivery of each week's content will be in-person at the classroom listed above. When advised or appicated, content may be delivered online.

complete their weekly assignments and scheduled assessments throughout the semester.

When applicable, students may be able to access online lecture videos for some live class sessions. If available, the online lectures can be used should anyone be unable to attend the class or to revisit topics covered in previous sessions.

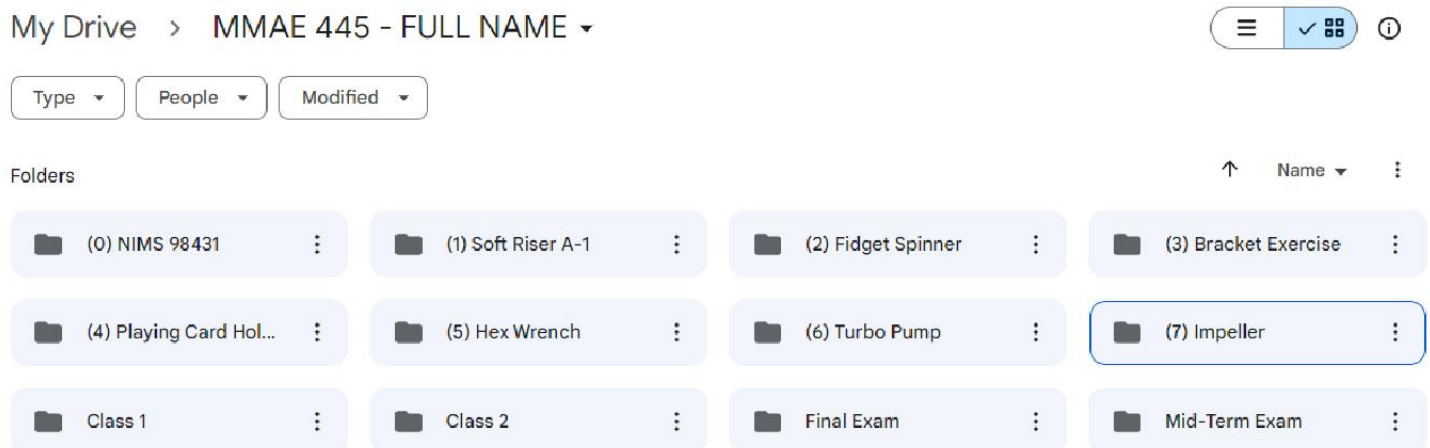
CLASS PARTICIPATION

Again - due to the hands-on nature of this class, the primary delivery of each week's content will be in-person at the classroom listed above. Participation is encouraged through in-class observation and/or the use of discussions, helping other students, etc.

This class was designed to explore many different aspects of the advanced manufacturing and computer aided engineering process. As a result, we will be relying on information provided to us from industry leaders, as well as case studies and real-world examples. **No formal textbooks are required.** All class material will be provided by the instructors.

Furthermore, each class session will have a folder for weekly assignments, lecture notes, training resources, and relevant handouts prior to each corresponding class session unless instructed otherwise. Any assignments must

EXERCISES, HOMEWORK, EXTRA CREDIT, AND SUBMISSIONS



Each project will have in-class exercise components and homework requirements.
Be sure to turn in everything listed in the instructions PDF found in each project's folder.

ALL EXTRA CREDIT MUST BE SUBMITTED USING THE FOLLOWING FORMAT:

- 1.) INCLUDE A SELFIE OF YOURSELF WHILE ATTENDING THE EVENT**
- 2.) PROVIDE A ONE PARAGRAPH SUMMARY OF YOUR EXPERIENCE**



GRADING

All Assignments will be graded based on content, the quality of work, class participation and their timely completion. Students are required to participate in the course discussion forum by responding to initial post and by interacting with fellow classmates as necessary. Points will be deducted for late or incomplete work. Course assessments will be weighted as follows:

In-Class Exercises	10%
Assignments	15%
Quizzes	15%
Midterm Exam	30%
Final Exam	30%
Quiz	15%

COURSE GRADING SCALE

MMAE 445	
Percentage (%)	Letter Grade
90-100	A
80-89	B
70-79	C
60-69	D
Below 60	E

STUDENT DELIVERABLES

The course design is to be participatory and interactive. A variety of instructional methods will be used in the course, including class discussions, case studies, videos, and presentations. Students will be expected to:

1. Complete the readings, training videos, and class assignments.
2. Actively participate in class discussions, discussion boards, and projects.
3. Complete the midterm and final examinations.

Students registered in the Mies/Main Campus course section are expected to attend weekly lectures. Each student has a professional obligation to participate in discussions unless prevented from doing so because of an illness or extenuating personal circumstances. Failure to complete all assignments and discussions will compromise the participant's mastery of the material, and adversely affect his or her performance in the course. Announcements posted in Blackboard or sent via email may include changes in assignment dates and/or content, changes in reading assignments or other important material. Unawareness of an Announcement posted in Blackboard will not be an acceptable excuse for any failure to meet course requirements.

ASSESSMENT CRITERIA FOR CLASS ASSIGNMENTS AND ACTIVITIES**1. In-Class Exercises (10%)**

Complete the in-class exercises in CAD and CAM for each in-person class session = 100 points
Arrangements will be made for Internet-only students to complete in-class assignments.

2. Assignments (15%)

Complete assignments as designated by the instructors on a weekly basis = 150 points
Arrangements will be made for Internet-only students to complete in-class assignments.

3. Quizzes (15%)

1/5 – 3D Model (CAD) = 15 points (x2 quizzes = 30 points total)

1/5 – FEA Study (CAD) = 15 points (x2 quizzes = 30 points total)

1/5 – Mechanical Assembly (CAD) = 15 points (x2 quizzes = 30 points total)

1/5 – 2D Blueprint (CAD) = 15 points (x2 quizzes = 30 points total)

1/5 – NC & G-Code Program (CAM) = 15 points (x2 quizzes = 30 points total) --- (= 150 points combined total)
Arrangements will be made for Internet-only students to complete the quizzes on a case-by-case basis.

4. Midterm Exam (30%)

1/5 – 3D Model (CAD) = 60 points

1/5 – FEA Study (CAD) = 60 points

1/5 – Mechanical Assembly (CAD) = 60 points

1/5 – 2D Blueprint (CAD) = 60 points

1/5 – NC & G-Code Program (CAM) = 60 points --- (= 300 points combined total)
Arrangements will be made for Internet-only students to complete the Midterm Exam on a case-by-case basis.

5. Final Exam (30%)

1/5 – 3D Model (CAD) = 60 points

1/5 – FEA Study (CAD) = 60 points

1/5 – Mechanical Assembly (CAD) = 60 points

1/5 – 2D Blueprint (CAD) = 60 points

1/5 – NC & G-Code Program (CAM) = 60 points --- (= 300 points combined total)
Arrangements will be made for Internet-only students to complete the Final Exam on a case-by-case basis.

= Total of 1,000 points or 100%

GRADING RUBRIC

	Poor = 0 pts Unsatisfactory <76%	Fair 76% – 84%	Good to Excellent 85% – 100%
Participation 10% of Course Grade	Not turned in = 0 pts.	Turned in on time, does not make reference connections to class content or reading.	Includes meaningful connections to class content or reading and responds to one other classmate.
IN- AND OUT-OF-CLASS ASSIGNMENTS 15% of Course Grade	Not turned in = 0 pts.	Turned in on time, shows minimal understanding of the concepts covered per deliverable. (Partial Credit) Points earned contribute to total overall grade.	Displays strong content knowledge by including relevant material and providing clear examples. Participates with (in-group) projects when applicable. (Full Credit) Points earned contribute to total overall grade.
QUIZZES 15% of Course Grade	Not turned in = 0 pts.	Turned in on time, shows minimal understanding of the concepts covered per deliverable. (Partial Credit) Points earned contribute to total overall grade.	Displays strong content knowledge by including relevant material and providing clear examples. (Full Credit) Points earned contribute to total overall grade.
MIDTERM EXAM 30% of Course Grade	Not turned in = 0 pts.	Exam questions are of equal point value.	Exam questions are of equal point value.
FINAL EXAM 30% of Course Grade	Not turned in = 0 pts.	Exam questions are of equal point value.	Exam questions are of equal point value.

ACADEMIC HONESTY

The IIT policy on Academic Honesty will be strictly enforced (see [IIT Student Handbook](#)). Students are expected to do their own work in completing assignments and course examinations. Students who copy other people's work and submit it as their own are guilty of plagiarism. Plagiarism can result in expulsion from the course and a grade of "E" (Failing). Interaction, discussion and conferring with peers are encouraged, but each student is expected to complete their own graded assignments. The use of unauthorized assistance or answer sheets when taking quizzes or exams is prohibited. For written assignments, all excerpts of text, statistics, graphs, etc. from published works (books, journals, websites) must be appropriately identified as quotations and include reference to the original source (title, author, date, page/section/website address).

GENERATIVE ARTIFICIAL INTELLIGENCE

As mentioned above, and in compliance with the university's policy towards Academic Honesty, students are expected to do their own work in completing assignments and course examinations. The use of Generative Artificial Intelligence tools, such as ChatGPT, may only be used as reference in regards to writing structure or subject context. Any use of Generative Artificial Intelligence tools must be made known through disclaimers, citations, and proof reading before submission. The act of copying and pasting Generative Artificial Intelligence responses for assignments or course examinations will be considered plagiarism. If the use of Generative Artificial Intelligence tools replaces or compromises the integrity of a student's work, then this stands in direct violation of the outcomes listed in the Course Objectives.

DISABILITY RESOURCES

Reasonable accommodations will be made for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources. The Center for Disability Resources (CDR) is located in 3424 S. State St., Room 1C3-2 (first floor), telephone:

ILLINOIS TECH'S SEXUAL HARASSMENT AND DISCRIMINATION INFORMATION

Illinois Tech prohibits all sexual harassment, sexual misconduct, and gender discrimination by any member of our community. This includes harassment among students, staff, or faculty. Sexual harassment of a student by a faculty member or sexual harassment of an employee by a supervisor is particularly serious. Such conduct may easily create an intimidating, hostile, or offensive environment.

Illinois Tech encourages anyone experiencing sexual harassment or sexual misconduct to speak with the Office of Title IX Compliance for information on support options and the resolution process.

You can report sexual harassment electronically at [iit.edu/incidentreport](https://www.iit.edu/incidentreport), which may be completed anonymously. You may additionally report by contacting the Title IX Coordinator, Virginia Foster at

or the Deputy Title IX Coordinator at

contact a licensed practitioner in Illinois Tech's Student Health and Wellness Center at

For a comprehensive list of resources regarding counseling services, medical assistance, legal assistance and visa and immigration services, you can visit the Office of Title IX Compliance website at <https://www.iit.edu/title-ix/resources>.

SIGNED SYLLABUS

I, _____, have read the Syllabus for **MMAE 445:**
Computer-Aided Design and Manufacturing in its entirety and understand the course's schedule, format, expectations, deliverables, grading, etc. Furthermore, as a student of Illinois Institute of Technology I will comply with the [Student Handbook](#) as it pertains to the Codes of: Academic Honesty, Conduct, Discipline, Sexual Harassment, Policies, Regulations, and Procedures.

Signature _____

Date _____

Keep a copy of this document for your records.